

UNIT-5

SECURITY IN AD HOC AND SENSOR NETWORKS

DISTRIBUTED SYSTEMS SECURITY

Distributed Systems Security in Ad-Hoc Sensor Networks focuses on ensuring the integrity, confidentiality, availability, and authenticity of data and communication in distributed systems formed by wireless sensor networks (WSNs), especially in ad-hoc sensor networks. These networks consist of sensor nodes that are deployed in an unstructured and dynamic manner and communicate wirelessly to collect and disseminate data. Given the characteristics of ad-hoc sensor networks, such as limited resources (e.g., battery life, memory, processing power), dynamic topology, and reliance on wireless communication, ensuring security is a challenging task.

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- **Security Solutions for Ad-Hoc Sensor Networks :**
- **1. Key Management:**
- **Pre-distribution of keys:** Secure key distribution is a challenge due to the lack of infrastructure in sensor networks. Key management protocols like **LEAP** (Localized Encryption and Authentication Protocol) or **RKB** (Random Key Pre-distribution) are designed to enable secure communication even in the absence of a centralized server.
- **Public Key Cryptography:** In some networks, **Elliptic Curve Cryptography (ECC)** or **RSA** can be used for secure key exchange and authentication, though these methods need to be lightweight to work effectively in resource-constrained environments.

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- **2. Authentication Protocols:**
- **Lightweight Authentication:** Protocols such as **TESLA** (Timed Efficient Stream Loss-tolerant Authentication) or **PAKE** (Password-Authenticated Key Exchange) are designed to provide secure authentication while considering the limited computational power of sensor nodes.
- **Mutual Authentication:** Both the source and destination nodes authenticate each other before establishing communication. This is crucial for preventing impersonation or spoofing attacks.

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- **3.Secure Routing Protocols:**
- **Secure Ad-Hoc Routing Protocols:** Traditional routing protocols like **AODV** (Ad hoc On-Demand Distance Vector) or **DSR** (Dynamic Source Routing) can be enhanced to incorporate security features, such as encryption, integrity checks, and secure neighbor discovery, to protect against routing attacks (e.g., blackhole attacks).
- **SEAD** (Secure Efficient Distance Vector) and **ARAN** (Authenticated Routing for Ad-hoc Networks) are examples of secure routing protocols tailored for ad-hoc sensor networks.

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- **4. Intrusion Detection Systems (IDS):**
- **Intrusion detection systems** are employed to detect malicious activities within the network. These systems use anomaly-based or signature-based approaches to identify unusual behavior, such as abnormal data flows or unauthorized node activities.
- Distributed IDS systems are often used in sensor networks, where each node or subset of nodes is responsible for monitoring specific types of attacks.
- **5. Data Integrity and Authentication:**
- Data integrity can be ensured by using **hash functions, digital signatures, or message authentication codes (MACs)** to verify the authenticity of messages.
- **End-to-end encryption** ensures that even if data is intercepted, it remains unreadable to unauthorized parties.

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- **6.Energy-Aware Security:**
- Security mechanisms must be designed to be energy-efficient, considering that sensor nodes are battery-powered. Lightweight cryptographic algorithms (e.g., **Elliptic Curve Cryptography (ECC)**) and efficient key management techniques are critical to conserving energy while maintaining security.

SECURE ROUTING

- **Secure Routing in Ad-Hoc and Sensor Networks** is a critical area of research and development aimed at ensuring that communication in these networks is protected against various attacks and vulnerabilities. Both **ad-hoc networks** and **sensor networks** are inherently susceptible to various security challenges due to their decentralized nature, wireless communication, and resource constraints. Therefore, secure routing protocols are vital to protect the integrity, confidentiality, and availability of the network and the data transmitted across it.

SECURE ROUTING

- **Types of Attacks in Routing Protocols :**
- **1. Blackhole Attack:**
 - A malicious node in the network falsely advertises itself as having the shortest path to the destination and intercepts packets. The node then drops the packets instead of forwarding them, leading to data loss
- **2. Wormhole Attack:**
 - A pair of malicious nodes create a "tunnel" between them by relaying messages to each other over a long distance. This can mislead routing protocols and disrupt normal network operation.

SECURE ROUTING

- **3. Sybil Attack:**
 - A malicious node pretends to be multiple nodes, either to overwhelm a particular part of the network or manipulate routing protocols.
- **4. Route Spoofing:**
 - A malicious node advertises false or incorrect routing information to other nodes, causing the network to route packets incorrectly.
- **5. Denial-of-Service (DoS):**
 - Attackers may flood the network with malicious traffic, consume resources, and prevent legitimate communication from taking place.

SECURE ROUTING

- **Secure Routing Protocols for Ad-Hoc and Sensor Networks:**
- secure routing protocols aim to ensure the following objectives:
- **Confidentiality:** Ensuring that the data exchanged in the network cannot be intercepted or read by unauthorized nodes.
- **Integrity:** Ensuring that data cannot be altered or tampered with during transmission.
- **Authentication:** Ensuring that nodes are verified and that data is only sent between legitimate nodes.
- **Availability:** Ensuring that the network remains functional even in the presence of malicious attacks

SECURE ROUTING

- **SECURE ROUTING PROTOCOLS:**

- **1.** Secure AODV (Ad-Hoc On-Demand Distance Vector)
- **2.** SEAD (Secure Efficient Ad-Hoc Distance Vector)
- **3.** ARAN (Authenticated Routing for Ad-Hoc Networks)
- **4.** DSR (Dynamic Source Routing) with Security Enhancements
- **5.** TORA (Temporally Ordered Routing Algorithm) with Security
- **6.** LEACH (Low-Energy Adaptive Clustering Hierarchy) with Security
- **7.** BHP (Beacon-Based Hello Protocol)

SECURE ROUTING

- **Secure AODV (Ad-Hoc On-Demand Distance Vector):**
- **AODV** is a popular routing protocol used in ad-hoc networks, which builds routes on demand and only when needed. However, AODV alone does not provide any security features.
- **Secure AODV (SAODV)** is a modified version that incorporates cryptographic techniques to ensure the authenticity and integrity of the routing messages:
- **Digital signatures:** To ensure that routing control messages, such as Route Request (RREQ) and Route Reply (RREP), are coming from legitimate sources.
- **Authentication:** To ensure the legitimacy of the source node and to prevent unauthorized nodes from injecting false routing information.

SECURE ROUTING

- **SEAD (Secure Efficient Ad-Hoc Distance Vector):**
- **SEAD** is a secure routing protocol designed to provide security for the **Destination-Sequenced Distance Vector (DSDV)** routing protocol. It uses **cryptographic hash functions** to prevent malicious nodes from altering the routing tables.
- **Hash chains** are used to authenticate routing information and ensure the integrity of route updates.
- **Secure metrics** are used to prevent attackers from modifying the metrics of the routes.
- SEAD is particularly lightweight, making it suitable for sensor networks with limited resources.

SECURE ROUTING

- **ARAN (Authenticated Routing for Ad-Hoc Networks):**
- **ARAN** is a secure routing protocol that focuses on the authentication of routing messages to prevent attacks like route spoofing and message forgery.
- Every node has a **public/private key pair**, and **digital signatures** are used to authenticate messages and routes.
- **Certificate authorities** are used to distribute keys and certify nodes' public keys
- ARAN provides robust security guarantees but may introduce additional overhead due to public key operations.

SECURE ROUTING

- **DSR (Dynamic Source Routing) with Security Enhancements:**
- **DSR** is an on-demand routing protocol that allows a source node to determine the route to the destination dynamically. Security enhancements to DSR include:
- **Secure route discovery:** Ensuring that the route discovered by the source node is authentic and not altered by malicious nodes.
- **Route validation:** Each routing message is digitally signed by the source node to ensure that it has not been tampered with.
- **Secure acknowledgments:** To ensure that the route reply messages come from legitimate nodes and that the sender's identity cannot be spoofed.

SECURE ROUTING

- **TORA (Temporally Ordered Routing Algorithm) with Security:**
- **TORA** is a highly adaptive routing protocol used for mobile ad-hoc networks. It uses a **link reversal** algorithm to maintain routes.
- In a secure version of TORA, routing messages are authenticated using cryptographic techniques to prevent attacks such as **blackhole** and **wormhole**.
- TORA's hierarchical approach allows it to handle large networks while implementing security at each stage of route construction and maintenance.

SECURE ROUTING

- **LEACH (Low-Energy Adaptive Clustering Hierarchy) with Security:**
- **LEACH** is a popular clustering-based routing protocol for sensor networks. In LEACH, the network is divided into clusters, and one node (cluster head) is responsible for communication with other clusters and the sink node.
- **Secure LEACH** adds encryption and authentication to prevent attacks such as node impersonation and data tampering within clusters.
- It ensures that only legitimate cluster heads are allowed to communicate with the sink, using **secure key management** and **cryptographic functions** to protect the communication.

SECURE ROUTING

- **BHP (Beacon-Based Hello Protocol):**
- **BHP** is used in wireless sensor networks to detect and avoid attacks related to false hello messages, such as **Hello Flood attacks**. It relies on periodic beacon exchanges to verify the existence of neighboring nodes and their connectivity.
- By using **cryptographic authentication** to sign beacon messages, BHP ensures that only authorized nodes can participate in the network and prevents attackers from impersonating legitimate nodes.

COOPERATION IN MANETS

- **Cooperation in MANETs (Mobile Ad-Hoc Networks) and Sensor Networks** is a crucial aspect of ensuring the efficient functioning, reliability, and security of these decentralized, wireless networks. Since MANETs and sensor networks are typically composed of nodes that collaborate without centralized control, cooperation among the nodes is essential to maintain network connectivity, routing, data collection, and other collaborative tasks.

COOPERATION IN MANETS

- Cooperation is two types :
- 1.CONFIDANT
- 2.TOKEN-BASED

COOPERATION IN MANETS

- **CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS) :**
- In **dynamic ad-hoc networks** (MANETs), cooperation between nodes is crucial for the efficient operation of the network. These networks are typically composed of mobile nodes that communicate with each other without a centralized infrastructure. This makes cooperation among nodes even more important, especially since each node may act as a **router**, **sensor**, or **data source**, depending on the needs of the application.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **Cooperation in Dynamic Ad-Hoc Networks:**
- **1. Cooperative Routing:**
- **Cooperative routing protocols** ensure that nodes work together to forward packets. In a **MANET**, nodes must collaborate in selecting routes, maintaining routes, and transmitting data. In some cases, a node may not be able to send data directly to the destination and must rely on intermediate nodes to forward the data.
- **Cooperative AODV** (Ad-Hoc On-Demand Distance Vector) and **DSR** (Dynamic Source Routing) are examples of protocols where cooperation is key for route establishment. These protocols require nodes to share their routing information and forward packets on behalf of other nodes, which might not have a direct route to the destination.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **2. Energy-Efficient Cooperation:**
- **Energy conservation** is critical in ad-hoc networks, especially in sensor networks where nodes have limited power. Cooperative strategies often involve balancing the energy consumption among nodes to prevent some nodes from exhausting their battery too quickly.
- Techniques like **duty cycling** and **sleep-wake scheduling** help balance energy usage. Cooperative protocols can ensure that nodes that have sufficient energy participate more actively in relaying data.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **3. Data Aggregation and Collaboration:**
 - In sensor networks, nodes collaborate to gather and aggregate data before sending it to a sink node. This minimizes the amount of data that needs to be transmitted, saving energy and improving the overall efficiency of the network.
 - **Collaborative sensing** can also improve the quality of collected data, as multiple nodes work together to reduce errors in sensing and increase the reliability of data transmission.
- **4. Reputation-Based Systems:**
 - **Reputation systems** help ensure cooperation by evaluating the behavior of nodes over time. Nodes with a good reputation (i.e., those that forward packets successfully and do not drop data) are more likely to be trusted by other nodes.
 - A node's reputation can be dynamically adjusted based on its past performance in forwarding packets, responding to routing requests, or aggregating data. Nodes with poor reputations (those acting selfishly or maliciously) can be excluded from participating in network operations.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **5. Incentive-Based Cooperation:**
- **Incentive mechanisms** provide rewards for cooperative behavior. These can be in the form of credits, energy replenishment, or access to network resources. For example, nodes that forward packets for other nodes can receive credits that can be used to request services from other nodes.
- **Credit-based systems** or **game-theoretic models** can be used to ensure that nodes are incentivized to cooperate. In some models, nodes can earn credits for forwarding traffic and use them to request services (e.g., data forwarding) from others in the network.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **Fairness in Dynamic Ad-Hoc Networks:**
- **1. Fair Resource Allocation:**
- **Fairness** ensures that no single node monopolizes network resources (e.g., bandwidth, energy). In **multi-hop networks**, some nodes may have to relay traffic for others, which can create imbalances in energy consumption and bandwidth usage.
- **Fairness-based routing protocols** such as **FARA** (Fair Ad-Hoc Routing Algorithm) ensure that no node is disproportionately burdened by relaying data. By periodically rotating the responsibility for relaying traffic, fairness in energy consumption and resource usage is maintained.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- **2. Fairness in Data Forwarding:**
- Nodes in an ad-hoc network must fairly share the burden of relaying packets. **Cooperative forwarding schemes** ensure that all nodes contribute to packet forwarding, thus reducing the chance of selfish nodes monopolizing network resources.
- **Fairness metrics** are often introduced in these protocols to measure how much data each node forwards. Nodes that forward a disproportionate amount of data are considered more cooperative, while those that forward little data are penalized.
- **3.Fairness in Transmission Power:**
- **Transmission power control** can help ensure fairness in terms of energy usage. Some nodes may transmit at high power to reach distant nodes, while others may use lower power, saving energy.

CONFIDANT(COOPERATION OF NODES,FAIRNESS IN DYNAMIC AD HOC NETWORKS)

- Ensuring fairness involves ensuring that no single node consumes too much power, while others are left with insufficient resources to communicate. **Fair power control algorithms** regulate the transmission power levels of each node in a way that balances energy consumption across the network.
- **4. Traffic and Bandwidth Fairness:**
- Ensuring **fair bandwidth allocation** is important for applications that require reliable, timely communication (e.g., multimedia streaming). Nodes that send more data should not hog the available bandwidth at the expense of others.
- **Fair queuing** mechanisms can help manage bandwidth allocation, ensuring that each node receives a fair share of the network's bandwidth for sending its data.

TOKEN - BASED APPROACH IN COOPERATION IN MANETS

- **Token-based cooperation** in **MANETs** (Mobile Ad-Hoc Networks) is a mechanism designed to encourage nodes to participate in network tasks, such as relaying data, by using a token-based incentive system. This approach helps mitigate the problem of selfish behavior, where nodes may not be willing to forward packets or participate in network activities to conserve their own resources (like energy or bandwidth).
- In token-based cooperation, tokens act as **virtual currency** or **credits** that are distributed among nodes. These tokens can be earned by performing cooperative actions, such as forwarding data for other nodes or contributing to routing decisions. Conversely, nodes may need to spend tokens to access network services, like sending data through the network or requesting a route.

TOKEN - BASED APPROACH IN COOPERATION IN MANETS

- **Token-Based Cooperation :**
- **1. Tokens as Incentives:**
- Tokens are used as a form of **incentive** to encourage nodes to act cooperatively. Nodes earn tokens for forwarding data, helping other nodes with routing, or performing network management tasks.
- Nodes that act selfishly (i.e., refuse to forward data or cooperate) will eventually run out of tokens and may lose the ability to access network services, which incentivizes cooperative behavior.
- **2. Token Generation:**
- Tokens can be generated based on various factors such as the **amount of data transmitted**, the **number of packets forwarded**, or the **cooperation level** (e.g., a node's participation in routing and data forwarding).
- In some systems, **token rewards** are distributed after a node has successfully forwarded a certain number of packets or helped in establishing a route.

TOKEN - BASED APPROACH IN COOPERATION IN MANETS

- **3. Token Spending:**
 - Nodes need tokens to access network resources. For instance, when a node wants to send data through the network, it must spend a certain amount of tokens as a "payment" for the forwarding service provided by other nodes.
 - This mechanism ensures that **selfish nodes** are penalized for not contributing to the network, as they will eventually deplete their tokens and be unable to send data.
- **4. Token Redistribution:**
 - Tokens are **redistributed** among nodes over time based on their level of cooperation. If a node is consistently forwarding packets or participating in the network, it will accumulate tokens, which can be used for its future needs.
 - **Token redistribution** can also occur through a **fair distribution scheme**, where tokens are spread evenly across the network, ensuring all nodes have an opportunity to earn and spend tokens.

TOKEN - BASED APPROACH IN COOPERATION IN MANETS

- **5. Token Expiry:**
- Tokens can have an **expiry time** to prevent them from accumulating indefinitely. Expiring tokens ensure that nodes continuously contribute to the network to earn new tokens, rather than simply hoarding old ones.
- **Token-Based Cooperation Protocols:**
- **1.T-CODA (Token-based Cooperation for Ad-hoc Networks):**
- T-CODA is a **token-based cooperation protocol** for MANETs that distributes tokens among nodes based on their participation in data forwarding. Nodes earn tokens for forwarding packets and lose tokens for not cooperating. The earned tokens are used for sending data through the network.
- The protocol ensures that selfish nodes are penalized and that all nodes participate fairly, increasing overall network performance.

TOKEN - BASED APPROACH IN COOPERATION IN MANETS

- **2.TAP (Token-based Ad-hoc Protocol):**
- TAP is another token-based protocol designed to encourage cooperative behavior in MANETs. Each node in the network earns tokens based on its contribution to the network, and nodes must spend tokens to request routing services or send data. TAP uses a **fair sharing** approach to ensure that nodes with fewer resources still have a chance to participate in network communication.

WSN SECURITY

- **Wireless Sensor Networks (WSNs)** are critical in various applications, ranging from environmental monitoring and healthcare to military and industrial applications.
- **Security Mechanisms for WSNs in Ad-Hoc Networks :**
- **1. Cryptographic Techniques:**
- **Symmetric and Asymmetric Encryption:** WSNs use cryptographic algorithms like **AES** (Advanced Encryption Standard) or **RSA** to ensure confidentiality and integrity. However, because of the computational overhead, lightweight encryption methods such as **Elliptic Curve Cryptography (ECC)** are becoming more popular
- **Hashing** and **MACs** are also commonly used for verifying the integrity of data.

WSN SECURITY

- **2. Key Management:**
- In WSNs, **key management** is crucial since nodes lack a central authority. **Key distribution schemes** like **pre-deployed keys**, **pairwise keys**, and **public key infrastructure (PKI)** are used to enable secure communication between nodes.
- **Key agreement protocols** such as **Diffie-Hellman** or **Elliptic Curve Diffie-Hellman (ECDH)** are used to establish secure communication channels between nodes.
- **3. Secure Routing Protocols:**
- **Secure routing protocols** like **SEAD** (Secure Efficient Ad-hoc Distance Vector), **AODV**, and **DSR** have been modified for WSNs to address the risks of attacks like **black hole**, **wormhole**, and **Sybil** attacks.
- These protocols incorporate **cryptographic techniques** to validate routing information and ensure that only legitimate nodes participate in the routing process.

WSN SECURITY

- **4. Intrusion Detection Systems (IDS):**
 - IDS are deployed to monitor network traffic and detect malicious activities such as **unauthorized access** or **DoS attacks**. These systems use various techniques like **anomaly detection**, **signature-based detection**, and **behavioral analysis**.
- **5. Data Aggregation and Authentication:**
 - In sensor networks, **data aggregation** reduces the amount of data transmitted, saving energy and bandwidth. Secure aggregation protocols ensure that data remains **confidential** and **intact** during the aggregation process.
 - **Secure data fusion** techniques ensure that aggregated data is verified for correctness and authenticity before it is used by decision-making systems.

WSN SECURITY

- **6.Trust Management Systems:**
- **Trust management** protocols assign trust values to nodes based on their previous behaviors. Nodes with high trust are more likely to be chosen for routing or data forwarding. If a node behaves maliciously (e.g., dropping packets), its trust value decreases, making it less likely to participate in network operations.
- **7. Cooperative Security:**
- **Cooperative security** models involve nodes working together to ensure network integrity. This can include cooperative **anti-jamming techniques** or mutual monitoring between nodes to detect abnormal behaviors.
- These approaches help address issues like **selfishness** and **collaborative attacks** (where multiple nodes collaborate to disrupt the network).

INTEGRATING MANETS,WLANS AND CELLULAR NETWORKS

Ingredients of a heterogeneous architecture

- A **heterogeneous architecture** refers to a system or network that is composed of different types of components, each with distinct capabilities, characteristics, or technologies. These systems are designed to leverage the strengths of each component to achieve better performance, scalability, and functionality than a homogeneous system made up of identical components. Heterogeneous architectures are commonly found in various domains, including computing, networking, and sensor networks.

Ingredients of a heterogeneous architecture

- **1.Diverse Computational Units (Processors) :**
- **Central Processing Units (CPUs):** Traditional processors used in computing tasks.
- **Graphics Processing Units (GPUs):** Specialized processors used for parallel processing, typically used in graphics rendering or deep learning tasks.
- **Field-Programmable Gate Arrays (FPGAs):** Customizable hardware used for specialized processing tasks with high efficiency.
- **Application-Specific Integrated Circuits (ASICs):** Custom-designed circuits optimized for specific tasks.
- **Digital Signal Processors (DSPs):** Processors optimized for signal processing tasks, commonly used in audio, video, and communication systems.
- **Microcontrollers:** Small computing units used in embedded systems, often used in the Internet of Things (IoT) and sensor networks.

Ingredients of a heterogeneous architecture

- **2. Memory Hierarchy :**
- **Shared Memory:** In some heterogeneous systems, different computational units may share a common memory pool. Efficient memory management is key to ensuring that data is accessible to all components in a balanced way.
- **Private Memory:** Each processor or computational unit may have its own private memory for faster access to local data. Managing the movement of data between local and shared memory is a critical challenge.
- **High-Bandwidth Memory (HBM):** Used to provide high-speed access to large volumes of data, which is important for handling complex computational tasks in heterogeneous environments.

Ingredients of a heterogeneous architecture

- 3. Interconnection Networks :
- **Communication Interfaces:** Different components (CPUs, GPUs, FPGAs, etc.) must be able to communicate with each other. This communication can happen over various networks like **PCI Express (PCIe)**, **InfiniBand**, **Ethernet**, or specialized communication protocols.
- **High-Speed Buses:** These are responsible for connecting different units within the architecture. Examples include **NVIDIA NVLink**, **AMD Infinity Fabric**, or **Thunderbolt**.

Ingredients of a heterogeneous architecture

- **4. Operating System and Software Support**
- **Task Scheduling:** The system must intelligently allocate tasks to the appropriate processor based on the task's characteristics and the processor's strengths. This requires advanced task scheduling algorithms and software support.
- **Middleware:** Middleware helps bridge the gap between different types of hardware and software. It abstracts the complexity of the heterogeneous components, providing a unified interface for developers and applications.
- **Parallel Programming Models:** Software frameworks such as **CUDA** (for GPUs), **OpenCL**, or **MPI** (Message Passing Interface) allow applications to utilize the different processing units effectively by distributing tasks.
- **Virtualization:** In some architectures, **virtual machines** or **containers** can be used to efficiently manage resources across different computational units.

Ingredients of a heterogeneous architecture

- **Data Management and Storage:**
- **Distributed Storage:** In heterogeneous systems, different components may require access to large datasets. Storage systems must be designed to handle distributed data, such as **network-attached storage (NAS)** or **cloud storage**.
- **Data Consistency and Synchronization:** Ensuring that the data across different components is consistent and synchronized is a key challenge, particularly when data is being processed in parallel across heterogeneous units.

Ingredients of a heterogeneous architecture

- **Energy Efficiency Management:**
- **Power Management:** Different components in a heterogeneous system may have varying power consumption profiles. Managing the power usage to ensure energy efficiency is essential, especially in mobile and embedded systems.
- **Dynamic Voltage and Frequency Scaling (DVFS):** DVFS allows processors to adjust their power consumption based on the computational workload, which is often employed in heterogeneous architectures to balance energy efficiency and performance.
- Scalability
- **Horizontal Scalability:** This refers to the system's ability to scale by adding more heterogeneous components, like adding more GPUs or FPGAs to the system.

PROTOCOL STACK

- In a heterogeneous network environment, a **dual-mode mobile terminal** is capable of connecting to multiple types of networks, such as cellular (e.g., LTE, 5G) and Wi-Fi, depending on the conditions and the availability of the network. The protocol stack of such a device would need to accommodate the communication protocols for both types of networks, and may also involve a layer for managing the transition between these networks.

PROTOCOL STACK OF A DUAL – MODE MOBILE TERMINAL IN A HETEROGENEOUS NETWORK ENVIRONMENT

Application Layer (HTTP, FTP, VoIP, etc.)
Transport Layer (TCP, UDP, SCTP, etc.)
Session Layer (e.g., SIP for VoIP, etc.)
Network Layer (IP) (IPv4, IPv6)
Data Link Layer (Wi-Fi: IEEE 802.11, Cellular: LTE/5G)
Physical Layer (Wi-Fi Radio, LTE/5G Radio)

PROTOCOL STACK OF A DUAL – MODE MOBILE TERMINAL IN A HETEROGENEOUS NETWORK ENVIRONMENT

- **1. Application Layer:**
 - Contains applications such as VoIP, web browsing, file transfer, etc.
 - Typically interacts with both Wi-Fi and cellular networks transparently.
- **2. Transport Layer:**
 - Responsible for end-to-end communication between the terminal and other devices in the network.
 - Protocols like **TCP** (Transmission Control Protocol), **UDP** (User Datagram Protocol), and **SCTP** (Stream Control Transmission Protocol) are commonly used.
 - **Dual-mode terminals** may adapt based on the available network for optimization (e.g., using TCP over cellular and UDP over Wi-Fi for latency-sensitive applications).

PROTOCOL STACK OF A DUAL – MODE MOBILE TERMINAL IN A HETEROGENEOUS NETWORK ENVIRONMENT

- **3. Session Layer:**
 - Manages sessions between two devices, including establishing, maintaining, and terminating communication sessions.
 - For VoIP, protocols like **SIP (Session Initiation Protocol)** are used.
- **4. Network Layer (IP):**
 - Handles IP addressing, routing, and packet forwarding.
 - Both **IPv4** and **IPv6** could be supported by the terminal to ensure compatibility with different types of networks.
 - The device may be configured to select the appropriate network based on signal strength, load, or other factors.

PROTOCOL STACK OF A DUAL – MODE MOBILE TERMINAL IN A HETEROGENEOUS NETWORK ENVIRONMENT

- **5. Data Link Layer:**
 - For **Wi-Fi**, it uses IEEE 802.11 standards.
 - For **Cellular** networks, it can use standards like **LTE** (Long-Term Evolution) or **5G NR (New Radio)**.
 - The mobile terminal's network interface card (NIC) supports both types, allowing it to switch between the two.
- **6. Physical Layer:**
 - Responsible for the physical transmission of data over the wireless medium.
 - For Wi-Fi, this involves the radio frequency spectrum, while for LTE/5G, it involves different radio technologies (e.g., OFDM in LTE, NR in 5G).

PROTOCOL STACK OF A DUAL – MODE MOBILE TERMINAL IN A HETEROGENEOUS NETWORK ENVIRONMENT

- Example:
- In a scenario where the mobile terminal is in a location where both **Wi-Fi** and **LTE**(Long Term Evolution) networks are available:
 - 1. The terminal initially connects to **Wi-Fi** for high-speed data transmission.
 - 2. If the Wi-Fi signal becomes weak or drops, the terminal automatically switches to the **LTE** network for continued service.
 - 3. If the terminal is engaged in a **VoIP call** using **SIP** at the application layer, it needs to ensure that the call remains intact during the handover between networks.

COMPARISON OF THE INTEGRATED ARCHITECTURES

- In a heterogeneous network environment, **integrated architectures** for dual-mode mobile terminals enable seamless communication across multiple network types, such as **Wi-Fi** and **cellular networks** (e.g., LTE, 5G). These architectures aim to offer the best performance, flexibility, and seamless user experience by allowing mobile terminals to switch between networks depending on signal strength, cost, latency, or other factors.
- The integrated architectures can be broadly classified into **two types**
- **1. Non-Integrated Architecture**
- **2. Integrated Architecture.**

Non-Integrated Architecture

- The mobile terminal or device uses distinct protocols, components, or interfaces to handle communication over different networks, without direct integration or coordination.
- Features:
- **Separate Protocol Stacks:** Different protocol stacks are used for cellular and Wi-Fi networks, often resulting in a more compartmentalized approach.
- **Manual Switching:** The user or the system needs to manually select or configure which network to use based on the availability or preference.
- **No Seamless Handover:** Switching from one network to another may lead to disruption or call drops, as there is no smooth transition between networks.

Non-Integrated Architecture

- **Independence Between Networks:** The two networks operate independently, and there is no direct coordination between them for optimized performance.
- **Limited Load Balancing:** The network may not balance traffic effectively between the two available network interfaces.
- Example :

Use Case: A user at home with a mobile phone may manually switch between cellular data and Wi-Fi depending on which network provides better speed or lower cost.

Scenario: The user manually turns off cellular data when they are on a Wi-Fi network, and vice versa when leaving the Wi-Fi coverage area.

Integrated Architecture

- An integrated architecture involves a **unified management system** for both the cellular and Wi-Fi network interfaces, allowing the system to dynamically switch or combine networks as needed. This architecture enables a more seamless experience and higher efficiency.
- Features:
- **Unified Protocol Stack:** A single, integrated protocol stack manages both Wi-Fi and cellular communications. This allows the device to switch between networks without changing the underlying protocol or requiring manual intervention.
- **Seamless Handover:** The system can hand over active sessions (e.g., voice, data) from one network to another without dropping the connection. For example, if Wi-Fi signal strength drops, the terminal seamlessly switches to LTE or 5G without user impact.

Integrated Architecture

- **Network Selection and Switching:** The integrated system dynamically chooses the best network based on a combination of factors, such as signal strength, load, latency, or cost.
- **Load Balancing:** The system can distribute traffic between the available networks for optimal performance. For example, latency-sensitive applications (e.g., voice or video calls) may be assigned to the cellular network, while data-heavy applications (e.g., file downloads) are routed over Wi-Fi.
- **Hybrid Access:** Some integrated architectures allow simultaneous access to both Wi-Fi and cellular networks (e.g., **Wi-Fi + Cellular**), combining the strengths of both.

Integrated Architecture

- **Example :**
- **Use Case:** A user on a call in a Wi-Fi network moves to a cellular network without dropping the call. The system seamlessly transitions to the best available network.
- **Scenario:** A mobile user is downloading a large file while on Wi-Fi but switches to the cellular network to make a voice call. The device handles the transition and ensures the file download is not interrupted.